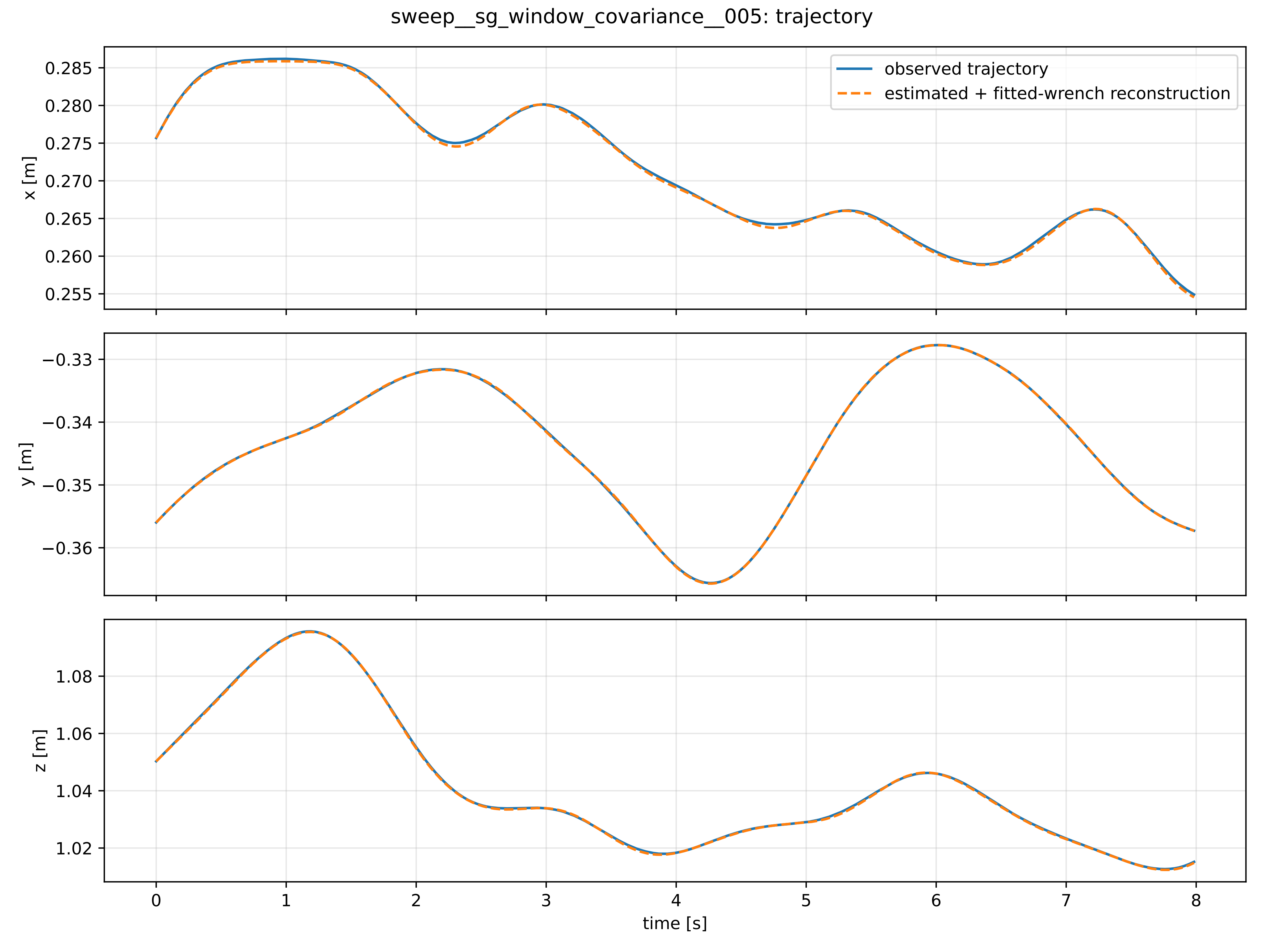
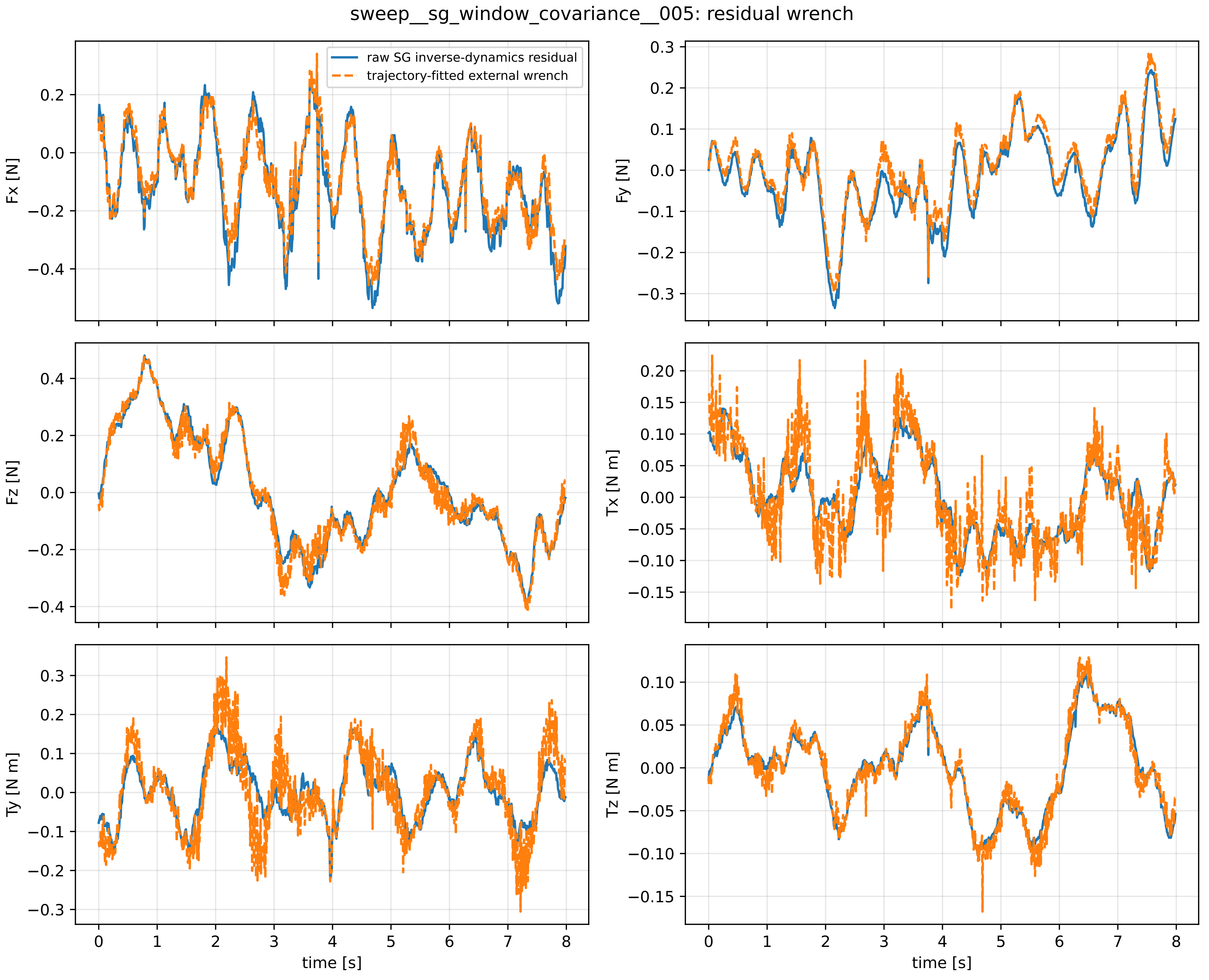
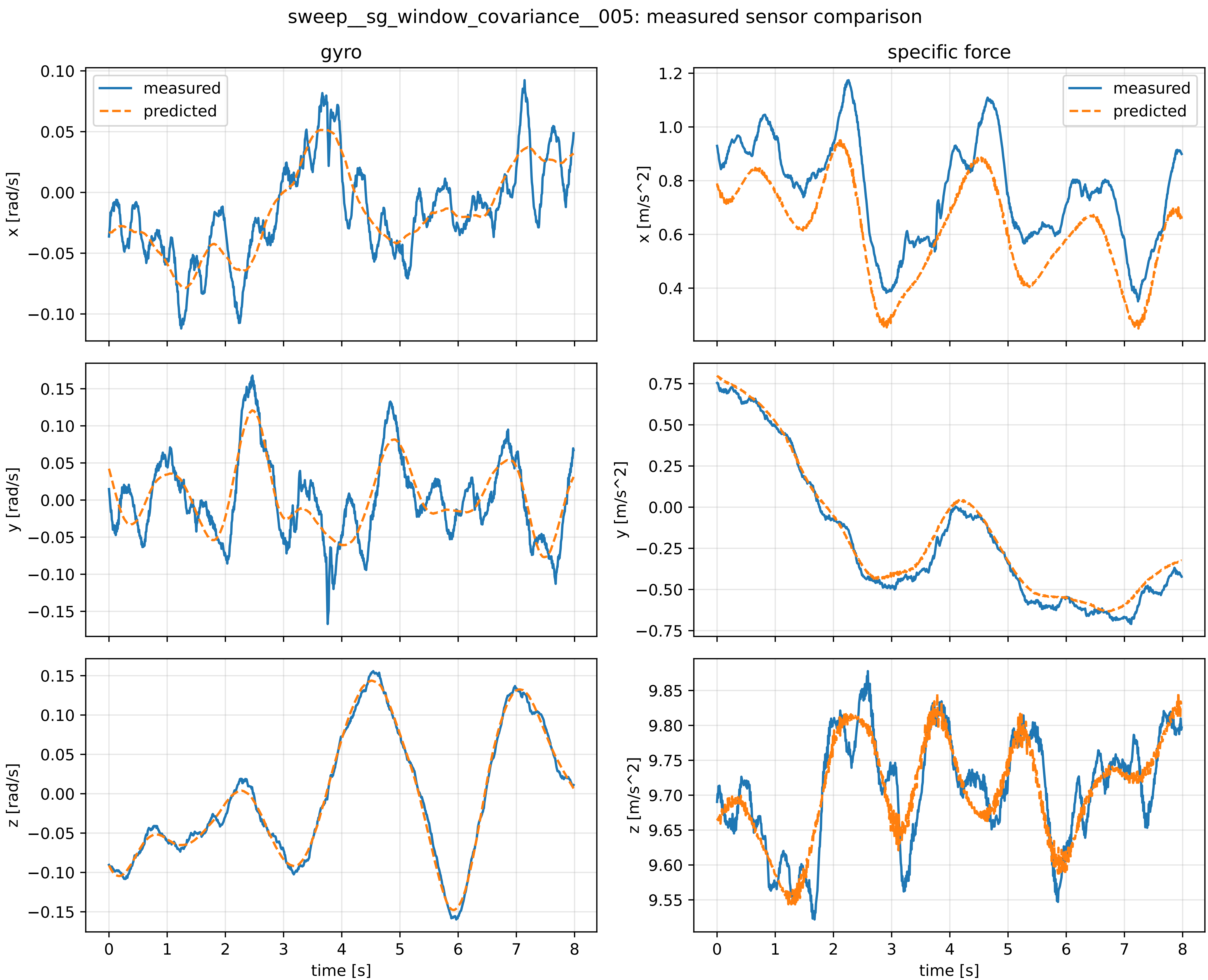

sweep__sg_window_covariance__005: trajectory
observed trajectory
estimated + fitted-wrench reconstruction
x [m]
0.285
0.280
0.275
0.270
0.265
0.260
0.255
y [m]
−0.33
−0.34
−0.35
−0.36
z [m]
1.08
1.06
1.04
1.02
0
1
2
3
4
5
6
7
8
time [s]



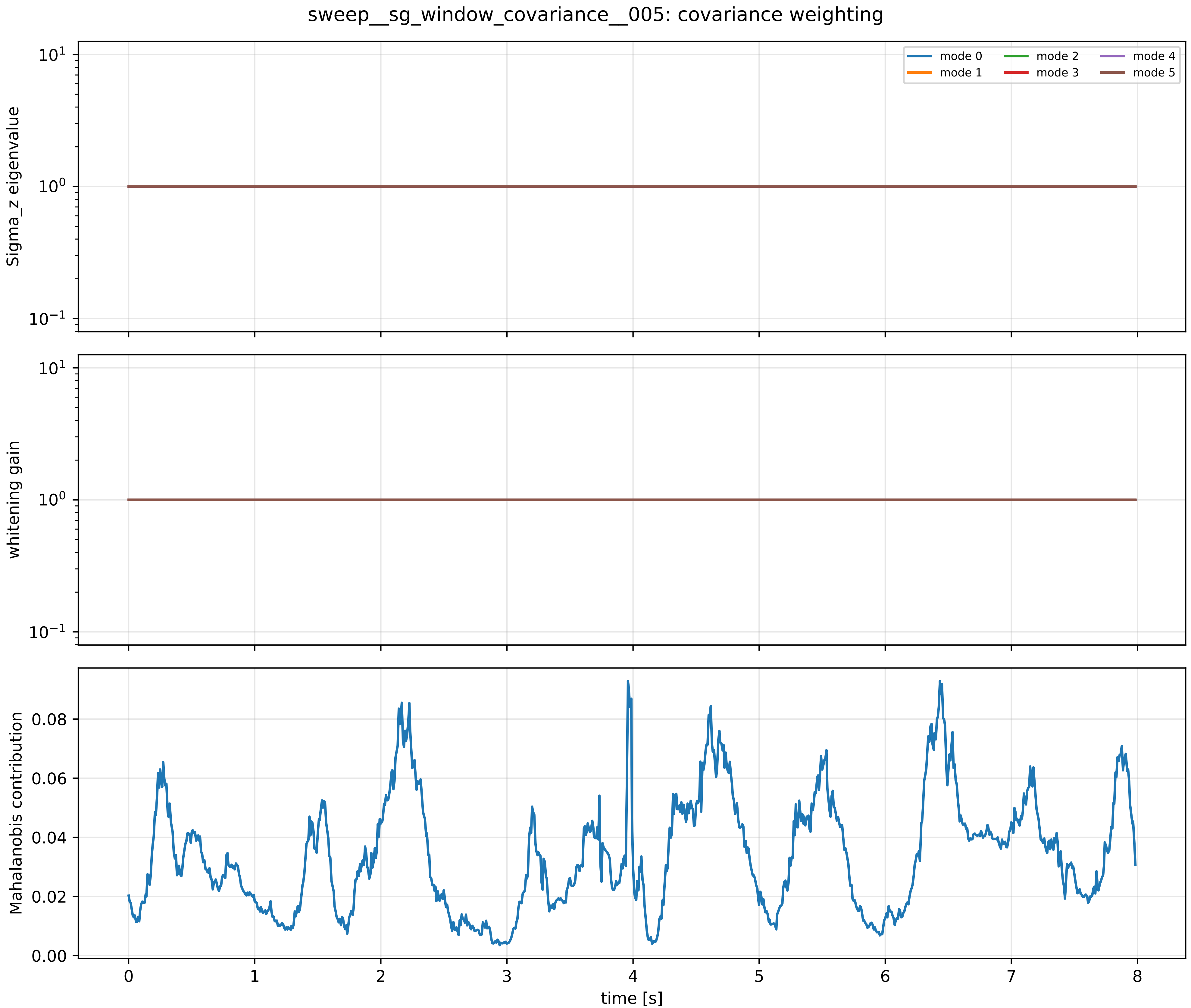
sweep__sg_window_covariance__005: residual wrench
raw SG inverse-dynamics residual
trajectory-fitted external wrench
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time [s]
time [s]



sweep__sg_window_covariance__005: measured sensor comparison
gyro
specific force
measured
predicted
x [rad/s]
y [rad/s]
z [rad/s]
x [m/s^2]
y [m/s^2]
z [m/s^2]
time [s]
time [s]

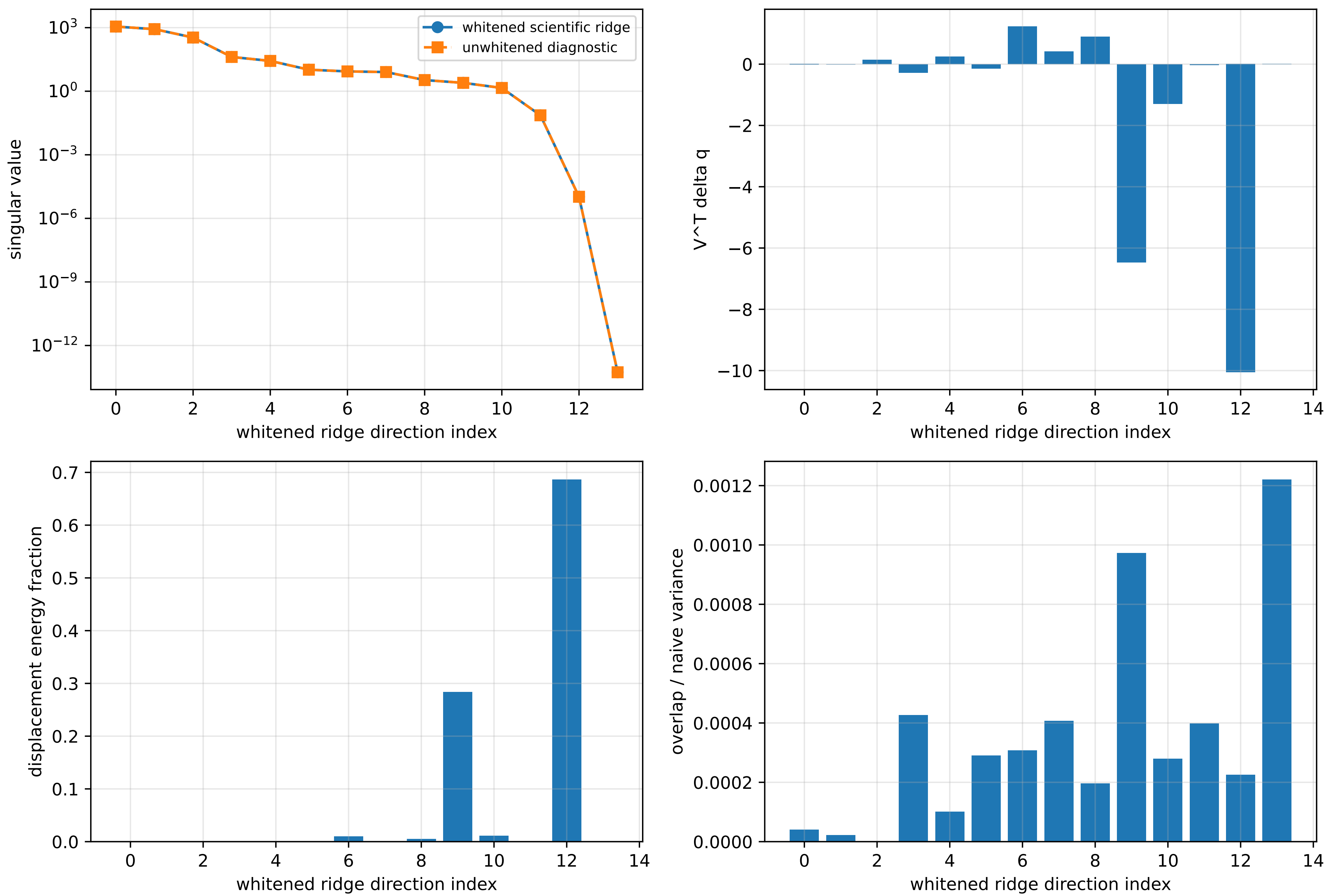
sweep__sg_window_covariance__005: nominal -> estimated parameters

```
Case: sweep__sg_window_covariance__005
status: completed (all stages completed)

mass [kg]       nominal  2.35159759   estimated  2.95600063   delta  0.604403034

inertia [kg m^2] (nominal -> estimated; delta)
  [ 6.5000061e-02 -7.2789925e-07  1.9015080e-05] -> [ 1.2041288 -0.0059258  0.1023283]   d=[ 1.1391288 -0.0059251  0.1023093]
  [-7.2789925e-07  6.4952656e-02  5.9167305e-08] -> [-0.0059258  0.8081026  0.0201984]   d=[-0.0059251  0.7431499  0.0201983]
  [1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [0.1023283 0.0201984 0.4224367]   d=[0.1023093 0.0201983 0.2934446]

CoG [m]       [-2.0247086e-03 -3.0526579e-05  9.5097496e-03] -> [-0.0164454 -0.0609444 -0.0144929]   d=[-0.0144206 -0.060913
rotor force effectiveness
  [1. 1. 1. 1.] -> [1.3155388 1.343913  0.6624044 0.7871958]   d=[ 0.3155388  0.343913  -0.3375956 -0.2128042]

rotor lag [s]  0.2
gimbal lag [s]  0.0500005484
```

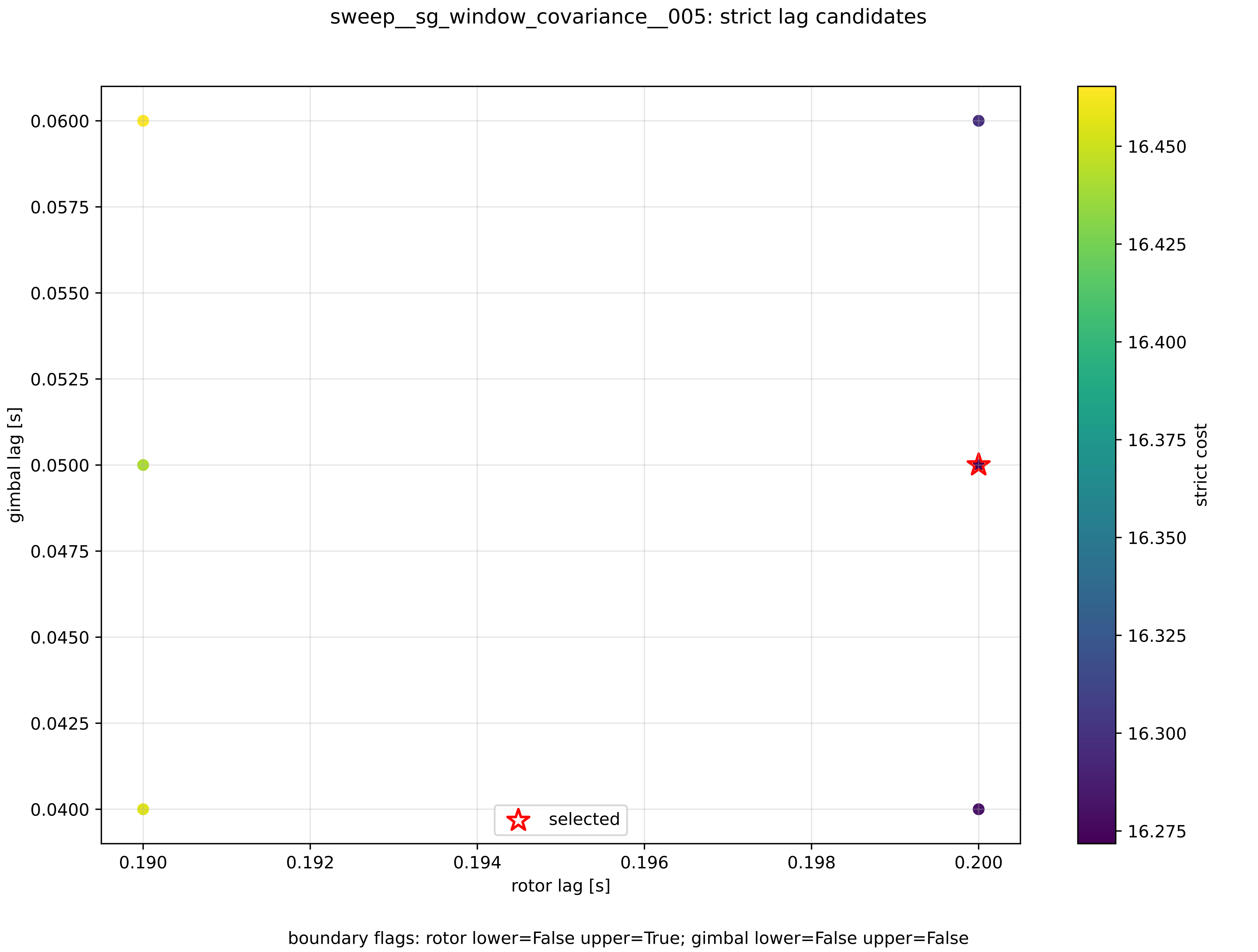


sweep__sg_window_covariance__005: covariance weighting
Sigma_z eigenvalue
whitening gain
Mahalanobis contribution
time [s]
mode 0
mode 1
mode 2
mode 3
mode 4
mode 5

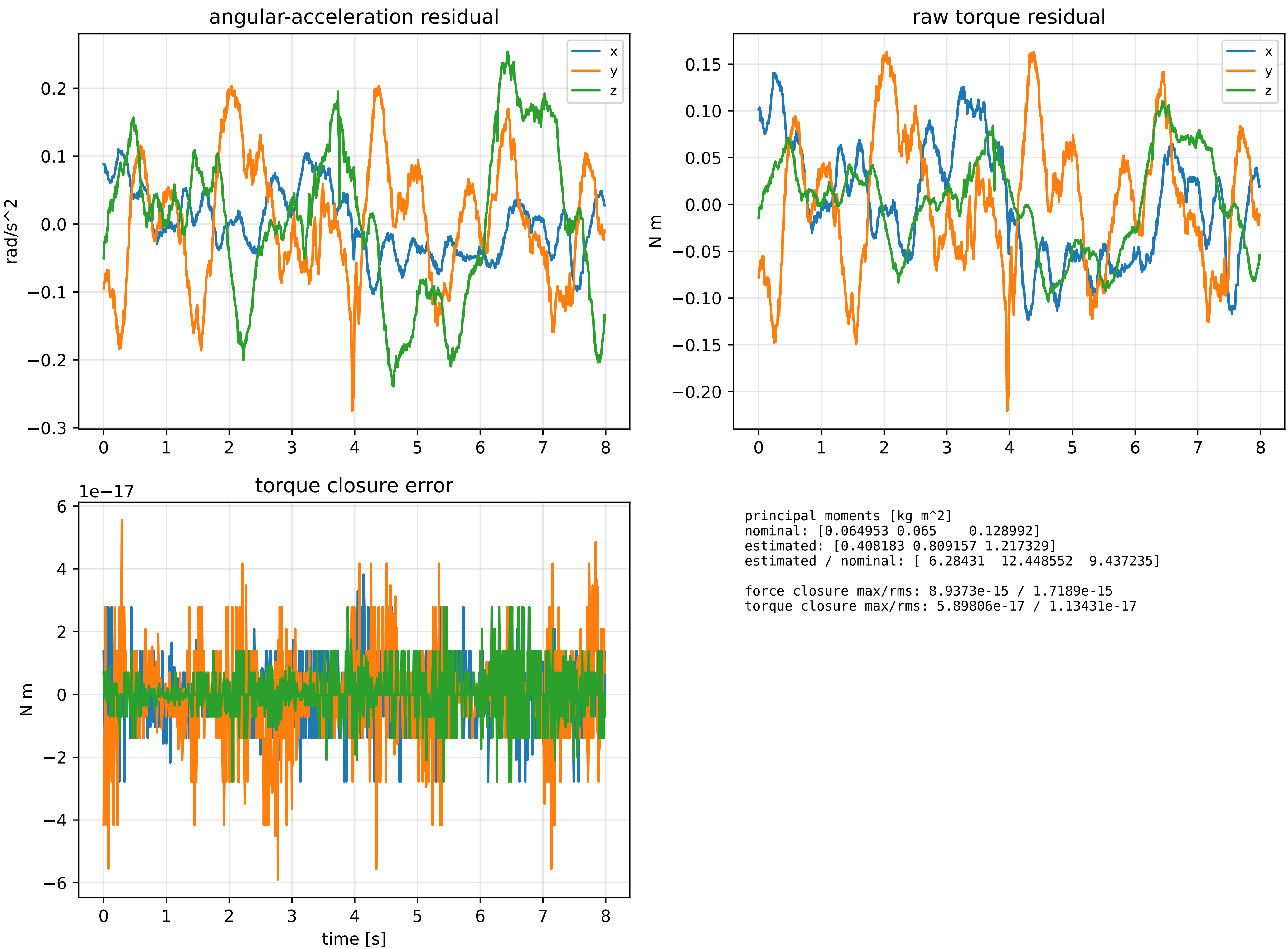
sweep__sg_window_covariance__005: parameter ridge



exact scale gauge: ||J v_scale||=5.09855e-14, rank=13, nullity=1


sweep__sg_window_covariance__005: strict lag candidates
gimbal lag [s]
0.0600
0.0575
0.0550
0.0525
0.0500
0.0475
0.0450
0.0425
0.0400
rotor lag [s]
0.190
0.192
0.194
0.196
0.198
0.200
selected
strict cost
16.450
16.425
16.400
16.375
16.350
16.325
16.300
16.275
boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False

sweep__sg_window_covariance__005: acceleration/wrench closure




```
principal moments [kg m^2]
nominal: [0.064953 0.065    0.128992]
estimated: [0.408183 0.809157 1.217329]
estimated / nominal: [ 6.28431  12.448552  9.437235]

force closure max/rms: 8.9373e-15 / 1.7189e-15
torque closure max/rms: 5.89806e-17 / 1.13431e-17
```